

Lab Sheet # L4

Instructions:

- Submit *.sql file, named as Lab04_6XXXXXX.sql to LAs or instructors prior to My Courses.
- If students cannot finish within the class time, students can submit until the end of the next class.

Self-study:

<https://www.mysqltutorial.org/mysql-basics/mysql-update/>

<https://www.mysqltutorial.org/mysql-basics/mysql-delete-join/>

<https://sentry.io/answers/in-mysql-how-do-i-fix-the-error-code-1175-during-an-update-in-mysql-workbench/>

Today's Task

0. Database Setup:
 - a. Import the Chinook database by executing the **L4_Chinook.sql** file.
(Don't need to submit)
1. Basic UPDATE
 - a. Write a query to update the name of a **customer** with **CustomerId = 3** to your

FirstName and **LastName**.

CustomerId	FirstName	LastName	Company	Address
1	Luís	Gonçalves	Embraer - Empresa Brasil...	Av. Brigadeiro Faria Lima,
2	Leonie	Köhler	NULL	Theodor-Heuss-Straße 34
3	Sir	Vac	NULL	1498 rue Bélanger
4	Bjørn	Hansen	NULL	Ullevålsveien 14

- b. Update the **Email** of the **customer** with **CustomerId = 5** to **itcs255@ict.com**.

CustomerId	FirstName	LastName	Email
1	Luís	Gonçalves	luisg@embraer.com.br
2	Leonie	Köhler	leonekohler@surfeu.de
3	Sir	Vac	ftremblay@gmail.com
4	Bjørn	Hansen	bjorn.hansen@yahoo.no
5	František	Wichterlová	itcs255@ict.com
6	Helena	Holý	hholy@gmail.com

- c. Update the **Address** of all customers in the "**Berlin**" to "**New Berlin**".

CustomerId	FirstName	LastName	Address	City
48	Johannes	Van der Berg	Lijnbaansgracht 120bg	Amsterdam
59	Puja	Srivastava	3,Raj Bhavan Road	Bangalore
38	Niklas	Schröder	New Berlin	Berlin
36	Hannah	Schneider	New Berlin	Berlin
42	Wuatt	Girard	9 Place Louis Barthou	Bordeaux

- d. Increase the **SupportRepId** of all customers by **1** where it is currently not **NULL**.

CustomerId	FirstName	LastName	SupportRepId
1	Luís	Gonçalves	4
2	Leonie	Köhler	6
3	Sir	Vac	4
4	Bjørn	Hansen	5

2. UPDATE with Join

- a. Write a query to update the **InvoiceDate** in the invoice table to the current date for all invoices belonging to customers from the **City** "Prague".

InvoiceId	CustomerId	InvoiceDate	City
77	5	2025-01-28 00:00:00	Prague
100	5	2025-01-28 00:00:00	Prague
122	5	2025-01-28 00:00:00	Prague
174	5	2025-01-28 00:00:00	Prague
295	5	2025-01-28 00:00:00	Prague

- b. Update the **BillingPostalCode** for invoices of customers from the city "**London**" to "**00000**".

InvoiceId	CustomerId	InvoiceDate	BillingPostalCode	City
11	52	2021-02-06 00:00:00	00000	London
140	52	2022-09-13 00:00:00	00000	London
163	52	2022-12-16 00:00:00	00000	London
185	52	2023-03-20 00:00:00	00000	London

- c. Set the **BillingCountry** of invoices for customers in "**Paris**" to "**France (Cap)**".

InvoiceId	CustomerId	BillingCountry	City
105	39	France (Cap)	Paris
128	39	France (Cap)	Paris
150	39	France (Cap)	Paris
202	39	France (Cap)	Paris

- d. Update the **Total** in the invoices table to increase by 10% for **customers** from "**Madrid**".

InvoiceId	CustomerId	Total	City
41	50	1.09	Madrid
162	50	2.18	Madrid
173	50	15.25	Madrid
228	50	9.80	Madrid

3. DELETE with Join

- a. Check the referenced relationship of **track** table by using the following script.

```

SELECT
    TABLE_NAME,
    COLUMN_NAME,
    CONSTRAINT_NAME,
    REFERENCED_TABLE_NAME,
    REFERENCED_COLUMN_NAME
FROM
    INFORMATION_SCHEMA.KEY_COLUMN_USAGE
WHERE
    REFERENCED_TABLE_NAME = 'table_name';

```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_T	REFERENCED_C
invoiceline	TrackId	FK_InvoiceLineTrackId	track	TrackId
playlisttrack	TrackId	FK_PlaylistTrackTrackId	track	TrackId

- b. Write a query to delete all tracks from the **track** table that belong to albums by the artist **Name** "AC/DC".

Note. You must delete the data from the child table (see result from a.) before delete from the parent table.

```

1 • SELECT COUNT(*) as TrackNum
2 FROM track t
3 JOIN album a ON t.AlbumId = a.AlbumId
4 JOIN artist ar ON a.ArtistId = ar.ArtistId
5 WHERE ar.Name = 'AC/DC';

```

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

TrackNum
0

- c. Delete all invoices for customers in the city "**Prague**".

Note. Follow the steps from a. and b. to check the relationship first.

```

34 • SELECT COUNT(*) as invoiceNum
35 FROM invoice i
36 JOIN customer c ON i.CustomerId = c.CustomerId
37 WHERE c.City = 'Prague';
38

```

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

invoiceNum
0

- d. Remove all **playlisttrack** records associated with **playlists** that contain tracks longer than 5 minutes.

```
SELECT COUNT(*) as playListNum
FROM playlisttrack pt
JOIN track t ON pt.TrackId = t.TrackId
WHERE t.Milliseconds > 300000;
```

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playListNum

0

- e. Delete all **customers** who have no invoices associated with them.

```
SELECT count(*) AS Null_customer
FROM customer c
LEFT JOIN invoice i ON c.CustomerId = i.CustomerId
WHERE i.InvoiceId IS NULL;
```

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Null_customer

0

4. ON DELETE CASCADE

- a. Check all the deleted rules and constraints names of the related table (given Artist is the parent table) by using the following script.

Note. ER is on last page.

```
SELECT
    kcu.TABLE_NAME,
    kcu.COLUMN_NAME,
    kcu.CONSTRAINT_NAME,
    kcu.REFERENCED_TABLE_NAME,
    kcu.REFERENCED_COLUMN_NAME,
    rc.DELETE_RULE
FROM
    INFORMATION_SCHEMA.REFERENTIAL_CONSTRAINTS rc
JOIN INFORMATION_SCHEMA.KEY_COLUMN_USAGE kcu
    ON rc.CONSTRAINT_NAME = kcu.CONSTRAINT_NAME
WHERE
    rc.CONSTRAINT_SCHEMA = 'database_name'
    AND kcu.REFERENCED_TABLE_NAME IN ('table_name',...);
```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_T	REFERENCED_C	DELETE_RULE
invoiceline	TrackId	FK_InvoiceLineTrackId	track	TrackId	NO ACTION
playlisttrack	TrackId	FK_PlaylistTrackTrackId	track	TrackId	NO ACTION
album	ArtistId	FK_AlbumArtistId	artist	ArtistId	NO ACTION
track	AlbumId	FK_TrackAlbumId	album	AlbumId	NO ACTION

- b. Add the foreign keys with ON DELETE CASCADE to the child table.

Note. You have to drop the existing foreign key first, then add the new foreign key with the ON DELETE CASCADE.

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_T/	REFERENCED_	DELETE_RULE
album	ArtistId	FK_AlbumArtistId	artist	ArtistId	CASCADE
invoice	TrackId	FK_InvoiceLineTrackId	track	TrackId	CASCADE
playlisttrack	TrackId	FK_PlaylistTrackTrackId	track	TrackId	CASCADE
track	AlbumId	FK_TrackAlbumId	album	AlbumId	CASCADE

- c. Delete the artist "Queen" and verify that all associated albums and tracks are deleted.

```

SELECT
  'Artist' AS TableName, COUNT(*) AS RecordCount
FROM artist
WHERE Name = 'Queen'
UNION ALL
SELECT
  'Albums', COUNT(*)
FROM album a
JOIN artist ar ON a.ArtistId = ar.ArtistId
WHERE ar.Name = 'Queen'
UNION ALL
SELECT
  'Tracks', COUNT(*)
FROM track t
JOIN album a ON t.AlbumId = a.AlbumId
JOIN artist ar ON a.ArtistId = ar.ArtistId
WHERE ar.Name = 'Queen'
UNION ALL
SELECT
  'InvoiceLines', COUNT(*)
FROM invoice il
JOIN track t ON il.TrackId = t.TrackId
JOIN album a ON t.AlbumId = a.AlbumId
JOIN artist ar ON a.ArtistId = ar.ArtistId
WHERE ar.Name = 'Queen'
UNION ALL
SELECT
  'PlaylistTracks', COUNT(*)
FROM playlisttrack pt
JOIN track t ON pt.TrackId = t.TrackId
JOIN album a ON t.AlbumId = a.AlbumId
JOIN artist ar ON a.ArtistId = ar.ArtistId
WHERE ar.Name = 'Queen';

```

TableName	RecordCount
Artist	0
Albums	0
Tracks	0
InvoiceLines	0
PlaylistTracks	0

Chinook ER

